# Shubham Kaushik

Ph.D. Researcher @ Brandeis University

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## RESEARCH INTERESTS

Databases, Data systems, Storage systems, Distributed systems, Data streaming, Cloud Data Platforms

## PROFESSIONAL EXPERIENCE

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| Jan 2024 - Present | **Ph.D. Researcher**<br>Brandeis University, MA, United States |
| Mar 2022 - Aug 2022 | **Software Engineer**, *Scalability & Infrastructure*<br>Kwalee, India |
| Jun 2021 - Mar 2022 | **Engineer - Information Security**, *Information Security Automation*<br>FIS Global, India |
| Oct 2019 - Jun 2021 | **Project Engineer**, *Cloud Computing*<br>Wipro Limited, India |
| Jul 2018 - Oct 2019 | **Project Engineer**, *Big Data Security*<br>Wipro Limited, India |
| Mar 2017 - Apr 2017 | **Full Stack Developer Intern**, *Full Stack Development*<br>SoPo Internet Private Limited, India |

## EDUCATION

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| Jan 2024 - Present | **Doctor of Philosophy (Ph.D.)** \| GPA: ***3.98/4.0***<br>Brandeis University, MA, United States<br>Major: **Computer Science** |
| Sep 2022 - Dec 2023 | **Master of Science (M.S.)** \| GPA: ***3.88/4.0***<br>Boston University, MA, United States<br>Major: **Computer Science** with specialization in *"Data-Centric Computing"* |
| Jul 2014 - Jun 2018 | **Bachelor of Technology (B.Tech.)**<br>Maharshi Dayanand University, Haryana, India<br>Major: **Computer Science & Engineering**<br>Thesis: *"Fault Modelling of an Object-Oriented System using Colored Petri Nets"* |

## PUBLICATIONS

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| **In Review** | Boao Chen, **Shubham Kaushik**, and Subhadeep Sarkar.<br>*Mind the Buffer: Dissecting the LSM-Buffer Design Space* [*Read Me*] |
| **ICDE 2026** | **Shubham Kaushik**, Manos Athanassoulis, and Subhadeep Sarkar. *RangeReduce: Query-Driven Compaction In LSM-Trees*, 42nd IEEE International Conference on Data Engineering [Talk] |
| **TPCTC 2025** | Alexander H. Ott, **Shubham Kaushik**, James Chen, and Subhadeep Sarkar. *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking*, TPC Technology Conference on Performance Evaluation & Benchmarking |
| **DBTest 2024** | **Shubham Kaushik** and Subhadeep Sarkar. *Anatomy of the LSM Memory Buffer: Insights & Implications*, In Proceedings of the International Workshop on Testing Database Systems |
| **JCSE 2019** | **Shubham Kaushik** and Ratneshwer. *Fault Modeling of an Object-Oriented System using CPN*, International Journal of Computer Sciences and Engineering |

## POSTERS

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| **NES Day 2026** | Alexander H. Ott, **Shubham Kaushik**, Boao Chen, and Subhadeep Sarkar *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking*, 2026 New England Systems Day |
| **NEDB Day 2025/2024** | **Shubham Kaushik**, Manos Athanassoulis, and Subhadeep Sarkar *RangeReduce: A Range Query Driven Compaction for LSM-Trees*, North East Database Day |

## TALKS

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| 05/2026 | "RangeReduce: Query-Driven LSM Compactions", ICDE'26, **Montréal**, Canada |
| 04/2026 | "RangeReduce: Query-Driven LSM Compactions", MiDAS Seminar, **Boston University**, USA |
| 01/2026 | "Tectonic: Bridging Sythentic and Real-World Workloads", 2026 New England Systems Day, **Harvard**, USA |
| 10/2025 | "Range Queries and LSM-Trees: From Foes to Friends", **Salesforce**, USA |
| 09/2025 | "Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking", **TPCTC**, UK |
| 06/2024 | "Anatomy of LSM Memory Buffers: Insights and Implications", **DBTest**, Santiago, Chile |

## TECHNICAL SKILLS

- **Programming Languages**: C, C++, Java, Python, SQL, Rust, Golang
- **Markup Languages**: HTML, CSS, JSON, YAML, LaTeX, Markdown
- **Databases**: RocksDB, Postgres, Redis, SQLite, ORM
- **Tools & Systems**: Linux, Asyncio, Git, Docker, Django, Django-REST, AWS, Apache Flink

## KEY PROJECTS

- **FluidLSM** (*Ongoing*): LSM-trees offer excellent write throughput but their performance is highly sensitive to configuration, which is a huge design space. Real workloads shift over time, and no static configuration remains optimal. FluidLSM adaptively reconfigure the LSM-tree on the fly, switching compaction strategies and tuning parameters in response to workload shifts, without manual intervention. [Blog]
- **Tectonic+** (*Ongoing*): A next-generation benchmarking suite designed to replace YCSB and other static workload generators for key-value stores. Tectonic+ extends the idea of Tectonic while adding multiple storage backends (RocksDB, Cassandra, Redis, ScyllaDB) through a unified database abstraction layer. [Blog] [GitHub]
- **Mind the Buffer** (*Under Review*): A comprehensive study of the LSM in-memory buffer design space, which is a small yet critical component of LSM-tree through which every read and write must pass. Evaluate nine buffer implementations across 1,200+ experiments, and present a practitioner handbook with 10 guidelines for buffer selection. Also, introduce an Adaptive buffer that monitors workload composition and switches implementations at flush boundaries, consistently matching the best under shifting workloads. [Blog] [GitHub]
- **RangeReduce** (*Published, ICDE 2026*): LSM trees suffer from poor range query performance due to high read amplification and redundant merging across sorted runs. RangeReduce piggybacks on range queries to compact data already read into memory, writing back only valid entries at a higher level — reducing read amplification and compaction overhead without adding a separate compaction pass. It achieves up to 18% lower range-query latency, 90% reduction in compaction debt, 12% less data movement, and 20% lower space utilization. [Paper PDF] [Blog] [GitHub]
- **Tectonic** (*Published, TPCTC 2025*): A Rust-based, highly configurable key-value workload generator that models the temporal, structural, and dynamic properties of real-world workloads, properties that YCSB and others cannot capture. Tectonic supports composite key generation, dynamic workload shifts, and user-specified data sortedness, achieving 2× higher throughput and 84% lower memory footprint than the state-of-the-art. [Paper PDF] [Blog] [GitHub]

## TEACHING ASSISTANT

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| Spring 2026/25/24 | *Database Management Systems* (*COSI 127B*), Brandeis University [SP '26] [SP '25] [SP '24] |
| Fall 2024/2025 | *Introduction to Computer Networking* (*COSI 128A*), Brandeis University |
| Fall/Spring 2023 | *Data Mechanics* (*DS 310*), Boston University |
| Fall 2022 | *Computer Networks* (*CS 455*), Boston University |

## PROFESSIONAL SERVICES

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| 2026 | Program Committee Member, VLDB 2027. |
| 2026 | Member, VLDB 2026 Shadow PC. |
| 2025 | Web Chair for *Northeast Database (NEDB) Day 2025*. |
| 2024/2025 | External reviewer for IEEE International Conference on Big Data. |
| 2024 - Present | Member, Association for Computing Machinery (ACM). |
| 2024 - Present | Student Member, Institute of Electrical and Electronics Engineers (IEEE). |

## CERTIFICATION

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| Jul 2023 | *"The Ultimate Hands-On **Hadoop**: Tame your **Big Data!**"* - Udemy [link] |
| Jul 2023 | *"Beginning **C++** programming from Beginner to Beyond"* - Udemy [link] |
| Oct 2018 | Statement of accomplishment for *"**Python** Track"* - DataCamp [link] |

## EXTRA CURRICULAR ACTIVITIES

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| Sep 2023/2025 | Judged and mentored at *HackMIT 2023*, aiding teams with technical challenges. |
| Nov 2022 | Mentored 4 teams, with an average of 20 participants at *BostonHacks*. |
| Jan 2017 | Volunteered in the Program Event Management team at the *National Youth Festival*. |